INTERPOLATION

A standard idea in interpolation is to find a polynomial $p_n(x)$ of degree n that assumes the given $n+1$ values; thus

$$p_n(x_0) = f_0, \qquad p_n(x_1) = f_1, \qquad \cdots, \qquad p_n(x_n) = f_n.$$

We call this $p_n(x)$ an interpolation polynomial and $x_0, \cdots, x_n$ the nodes. And if $f(x)$ is a mathematical function, we call $p_n(x)$ a polynomial approximation of f.

We use $p_n(x)$ to get the value of f for a specific x between x_0 to x_n ("interpolation") or sometimes outside this interval ("extrapolation").

continued

LAGRANGE INTERPOLATION

Lagrange interpolation polynomial is given by;

$$f_n(x) = \sum_{i=0}^{n} L_i(x) f(x_i) \qquad (1)$$

Where, n in $f_n(x)$ stands for n_{th} order polynomial that approximate the function $y = f(x)$ for $n+1$ number of data points, (x_0, y_0), (x_1, y_1) .. (x_n, y_n).

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j} \qquad (2)$$

$Li(x)$ is a weighting function that includes a product of n-1 terms omitting $j=i$ term.

LAGRANGE INTERPOLATION

Linear interpolation (2 points) is interpolation by the straight line through (x_0, f_0), (x_1, f_1); Thus the linear Lagrange polynomial p_1 is a sum $p_1 = L_0 f_0 + L_1 f_1$, Obviously,

$$L_0(x) = \frac{x - x_1}{x_0 - x_1}, \qquad L_1(x) = \frac{x - x_0}{x_1 - x_0}.$$

This gives the linear Lagrange polynomial

$$p_1(x) = L_0(x) f_0 + L_1(x) f_1 = \frac{x - x_1}{x_0 - x_1} \cdot f_0 + \frac{x - x_0}{x_1 - x_0} \cdot f_1. \qquad (3)$$

continued

QUADRATIC INTERPOLATION

The interpolation of 3 points given (x_0, f_0), (x_1, f_1), (x_2, f_2) by a second-degree polynomial $p_2(x)$, which by Lagrange's idea is

$$p_2(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2 \qquad (4a)$$

with $L_0(x_0) = 1$, $L_1(x_1) = 1$, $L_2(x_2) = 1$, and $L_0(x_1) = L_0(x_2) = 0$, etc. We claim that

$$L_0(x) = \frac{l_0(x)}{l_0(x_0)} = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$L_1(x) = \frac{l_1(x)}{l_1(x_1)} = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} \qquad (4b)$$

$$L_2(x) = \frac{l_2(x)}{l_2(x_2)} = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} .$$

continued

EXAMPLE1

Given (*9.0, 2.1972), (9.5, 2.2531*) and (*11.0, 2.3979*) are 3 data points. Compute the value of y at x= 9.2.

Solution. 3 points make a second-degree polynomial $p_2(x)$, which by Lagrange's idea is

$$p_2(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2$$

Where, $$L_0(x) = \frac{(x-9.5)(x-11.0)}{(9.0-9.5)(9.0-11.0)} = x^2 - 20.5x + 104.5, \qquad L_0(9.2) = 0.5400,$$

$$L_1(x) = \frac{(x-9.0)(x-11.0)}{(9.5-9.0)(9.5-11.0)} = -\frac{1}{0.75}(x^2 - 20x + 99), \qquad L_1(9.2) = 0.4800,$$

$$L_2(x) = \frac{(x-9.0)(x-9.5)}{(11.0-9.0)(11.0-9.5)} = \frac{1}{3}(x^2 - 18.5x + 85.5), \qquad L_2(9.2) = -0.0200,$$

Putting the values of L(x) in the above equation we get ,

$$p_2(9.2) = 0.5400 \cdot 2.1972 + 0.4800 \cdot 2.2513 - 0.0200 \cdot 2.3979 = 2.2192.$$

Example 2

Determine the value of *f(x)* or *y* at *x* =23 from the following table using Lagrange interpolation method.

x	**3**	**6**	**11**	**14**	**25**
y or *f*	1	10	171	93	15

[Remember 5 points produce 4th order polynomial

$$p_4(x) = L_0(x)\, f_o + L_1(x)\, f_1 + L_2(x)\, f_2 + L_3(x)\, f_3 + L_4(x) f_4$$

Where, $L_0(23) = \frac{(23-6)(23-11)(23-14)(23-25)}{(3-6)(3-11)(3-14)(3-25)} = \frac{(17)(12)(9)(-2)}{(-1)(-8)(-11)(-22)}$

$$= \frac{-3672}{1936} = -1.897$$

At the same way, calculate $\boldsymbol{L_1(23)}$ and so on